

Demonstration of Running a GWAS Pipeline via Gigwa and Galaxy

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1 Objective of the Demonstration

This documentation presents the complete execution of a GWAS analysis pipeline using a graphical interface. The workflow relies on using **Gigwa** for data preparation and export, followed by running the pipeline in **Galaxy**, and finally visualizing the results directly within **Galaxy**.

2 Data Preparation in Gigwa

2.1 Dataset Selection



FIGURE 1 – Gigwa Interface – project and dataset selection

The **Gigwa** interface allows selecting the genomic dataset to analyze. The chosen project contains genotype data as well as associated metadata.

2.2 Applying Filters on Variants

The screenshot shows the Gigwa v2.11 interface. At the top, there's a header with the Gigwa logo, a taxon dropdown set to '(Any taxon)', a species dropdown set to 'olea_europaea', a project dropdown set to 'Leccino', and an information icon. Below the header, on the left, is a sidebar with filter options: 'Variant types' (set to 'Any'), 'Sequences' (23/23, with a dropdown), 'Position (bp)' (with '≥' and '≤' input fields), 'Variant Effects' (set to 'Nothing selected'), 'Gene Names' (with 'Input Names here' and 'Selection mode' buttons), 'Filter by variant IDs' (with 'Input IDs here' and filter icons), and 'Investigate genotypes' (set to 'disabled'). At the bottom of the sidebar is a 'Work on samples' checkbox. On the right, there's a 'Search' button, an 'Enable browse and export' checkbox (checked), and a pagination bar showing '<' and '1 - 10'. Below this is a table of variants.

id	sequence	start	stop	alleles
GWHEUUU00000001.1		32704	32704	C A
GWHEUUU00000001.1		32714	32714	G A
GWHEUUU00000001.1		32727	32727	T C
GWHEUUU00000001.1		32767	32767	T C
GWHEUUU00000001.1		32794	32794	G A
GWHEUUU00000001.1		32811	32811	G A
GWHEUUU00000001.1		33380	33380	A T
GWHEUUU00000001.1		33395	33395	T G

FIGURE 2 – Defining filters and parameters in Gigwa

Filters can be applied to restrict the variants and individuals used for the analysis. This step ensures that only relevant data is exported to Galaxy.

2.3 Choosing the Phenotype and Associated Individuals

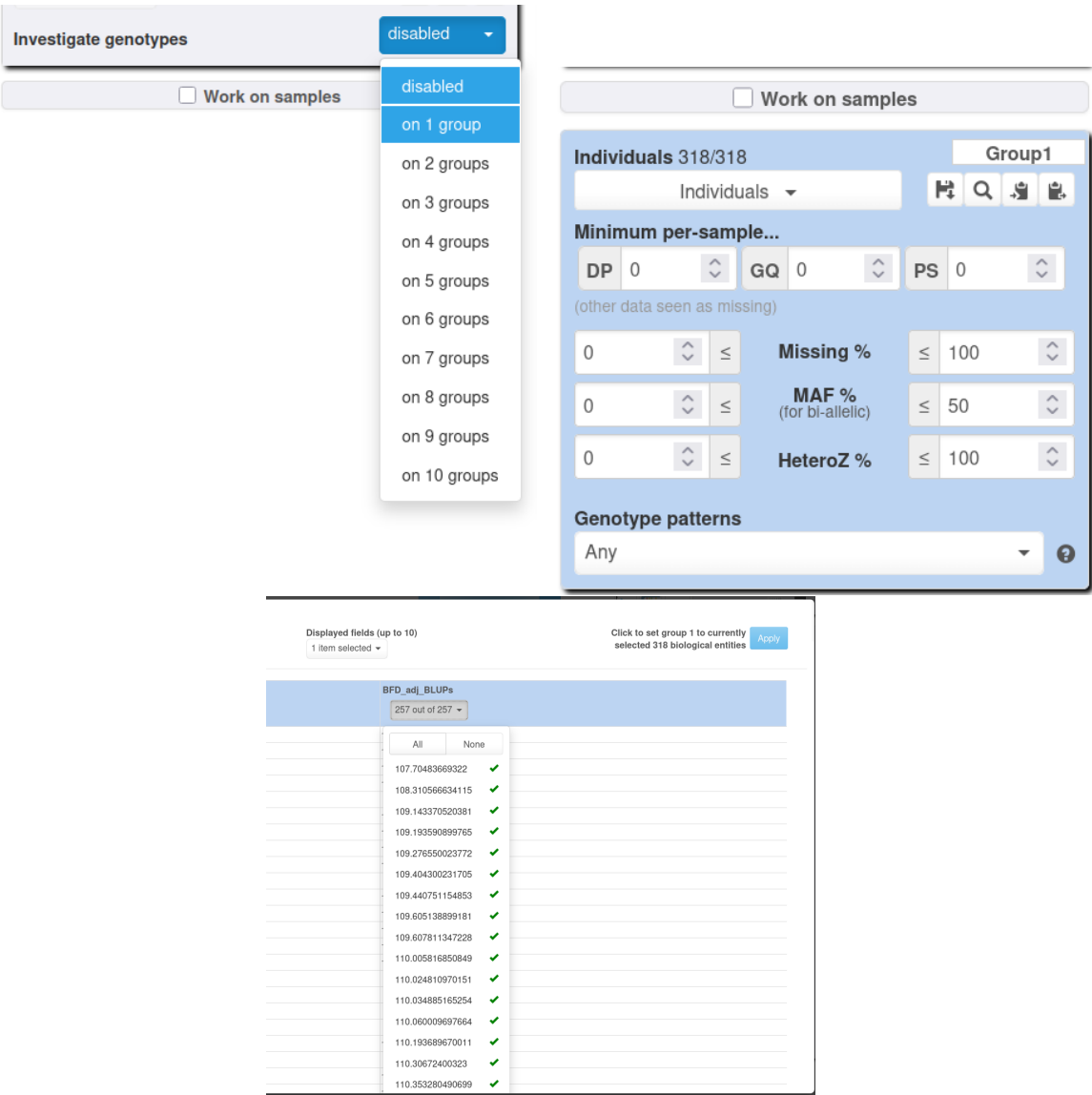


FIGURE 3 – Selecting the phenotype and individuals

To filter by individuals and select the phenotype, it is necessary to work with a single group, then click on the magnifying glass, select the individuals who have a value for the phenotype you want to work with, and finally click *Apply* to apply the filter.

2.4 Exporting Data to Galaxy

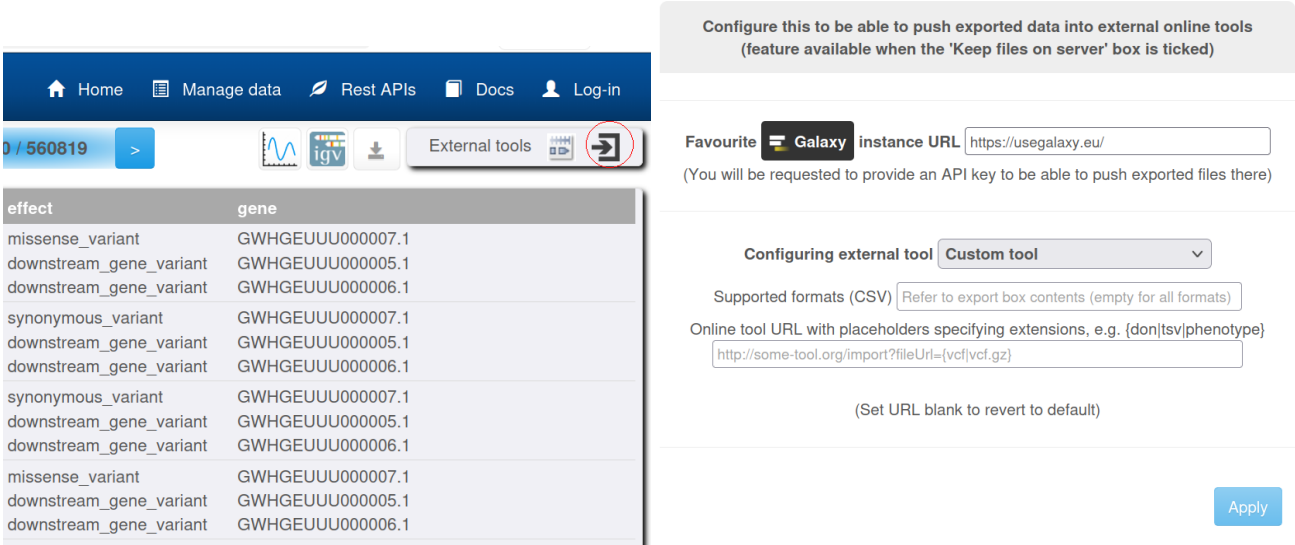


FIGURE 4 – Configuring the Galaxy instance

Before exporting data, it is necessary to define the Galaxy instance where the data will be sent. The pipeline is available at <https://usegalaxy.eu/>. First, click on the icon in the *External tools* section, then enter the URL <https://usegalaxy.eu/> in the *Favorite Galaxy instance URL* field.

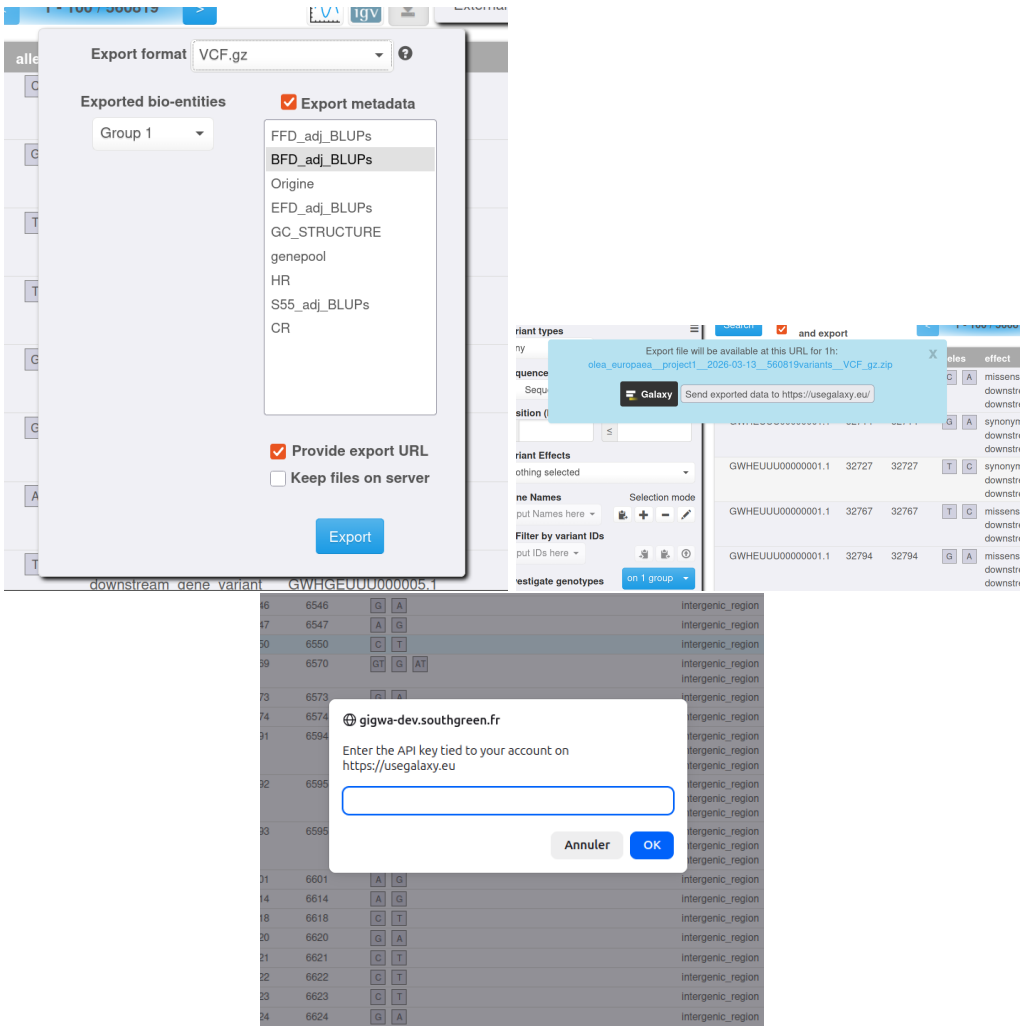


FIGURE 5 – Exporting data

To download the data and send it to Galaxy, click the download button, choose the VCF.gz format, select the

phenotype, check *Provide export URL*, and click *Export*. After the download completes, click the *Send exported data to* button. Gigwa will then ask for the API key associated with your Galaxy account.

2.4.1 How to Find the API Key Associated with Your Galaxy Account

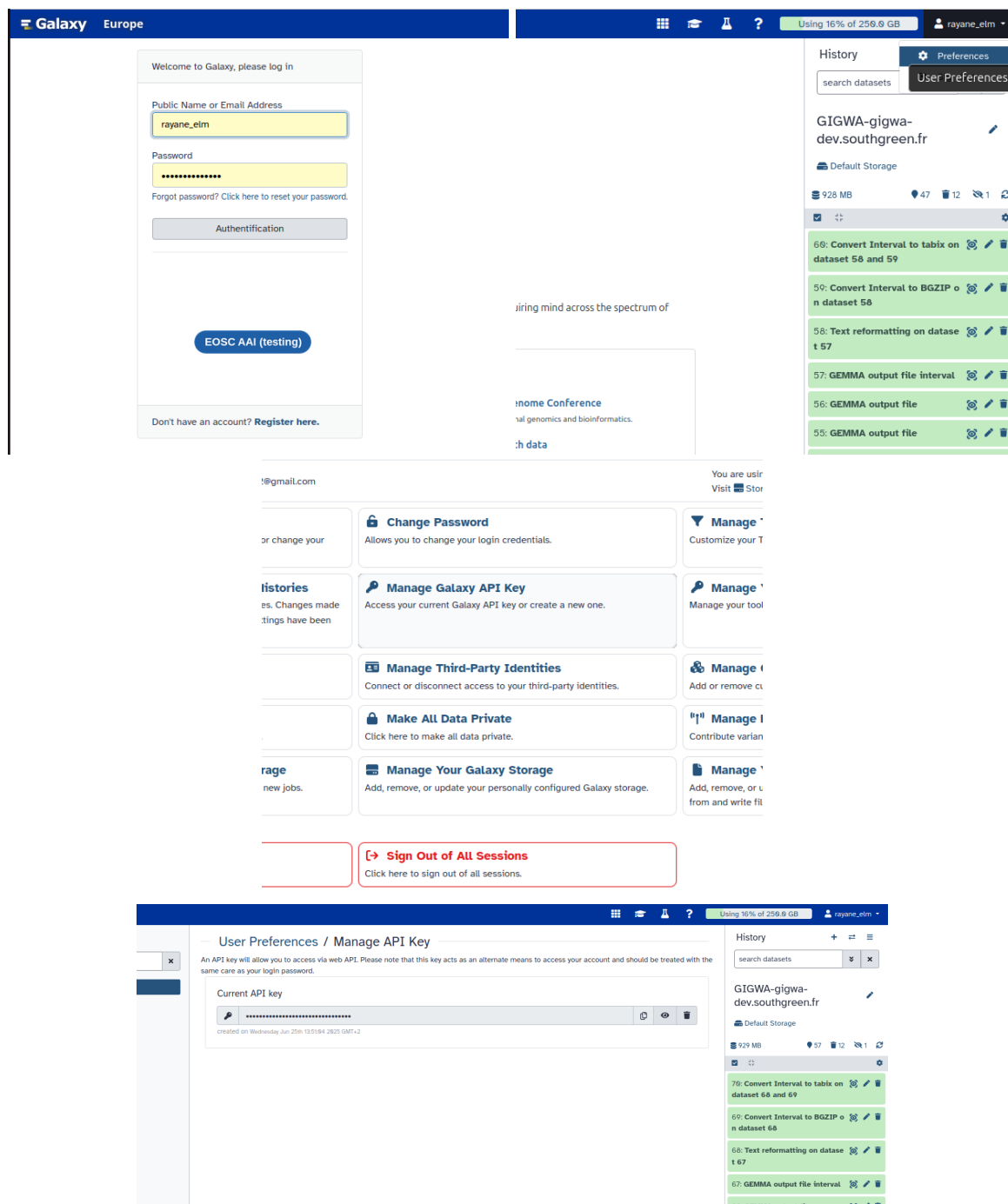


FIGURE 6 – Finding the API key

After logging into your Galaxy account, click on your username (top right), then click on Preferences, then on *Manage Galaxy API key* (if no key has been created yet, click *Create a new key*), and finally copy the API key and paste it into Gigwa. Two files (genotype in VCF.gz format and phenotype in TSV format) will be sent to Galaxy on the account associated with the API key and will appear in the GIGWA history. Procedure to change the history :

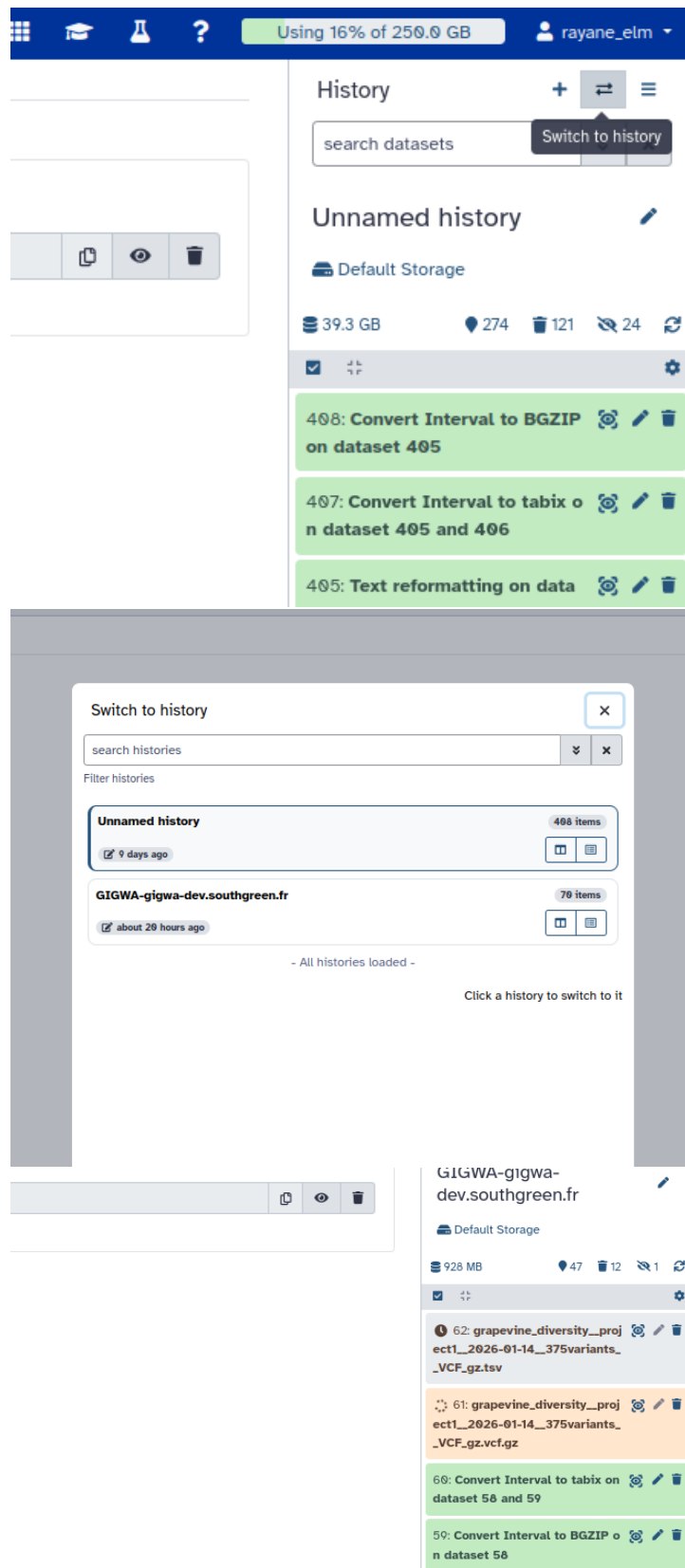


FIGURE 7 – Changing history in Galaxy

3 Running the Pipeline

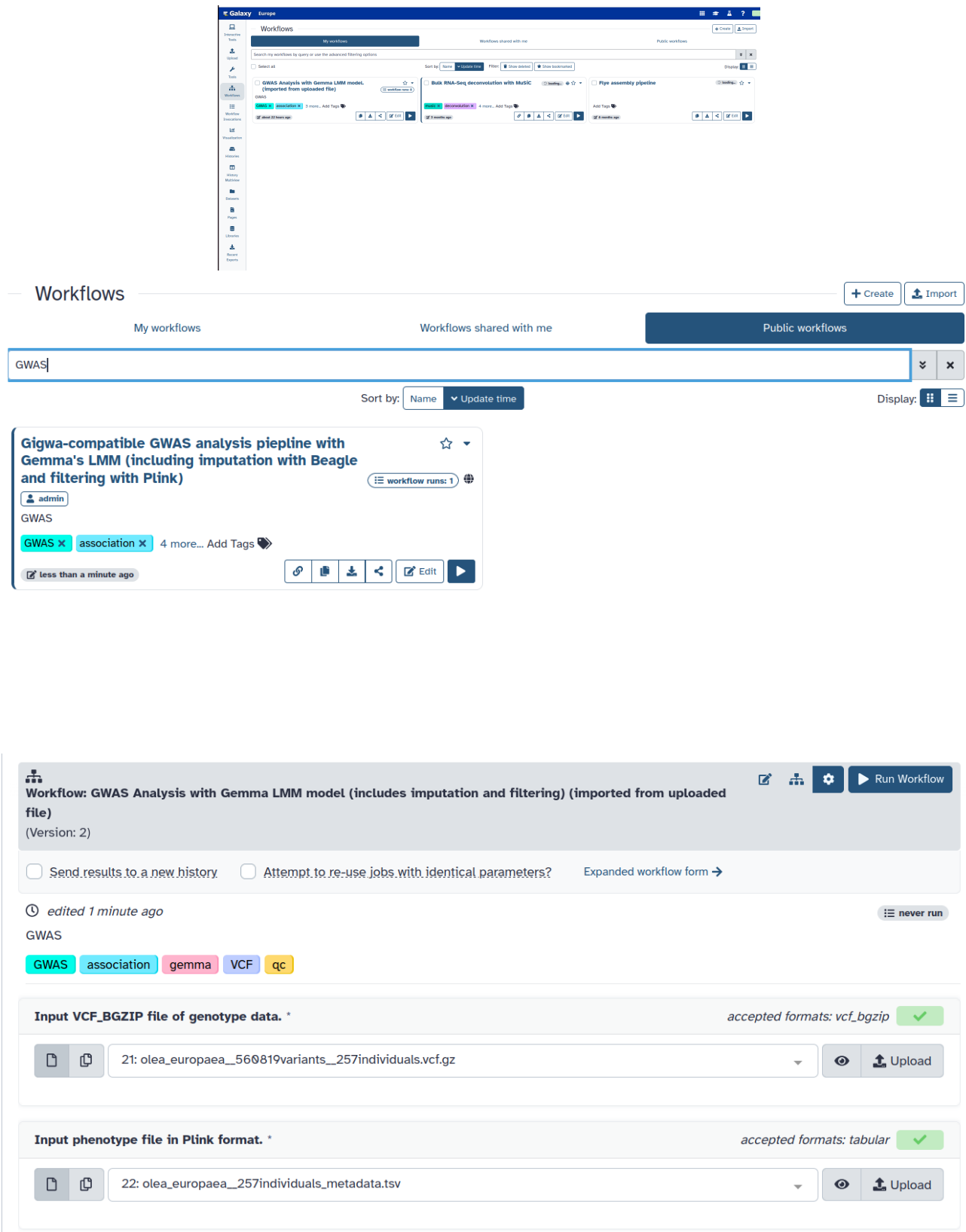


FIGURE 8 – Running the pipeline

Click on *Workflows* (left navigation bar), search in *Public workflows* for the pipeline by typing GWAS, click *Run workflow*, choose the two files (genotype in **vcf.gz** + phenotype in **tabular** format), click *Run workflow* and wait for it to complete.

4 Visualizing Results in Galaxy with LocusZoom

The last two files, **BGZIP** (format **gwas_bgzip**) and **Tabix** (format **gwas_tabix.gz**), are required to interactively visualize the GWAS analysis results with the integrated **LocusZoom** tool.

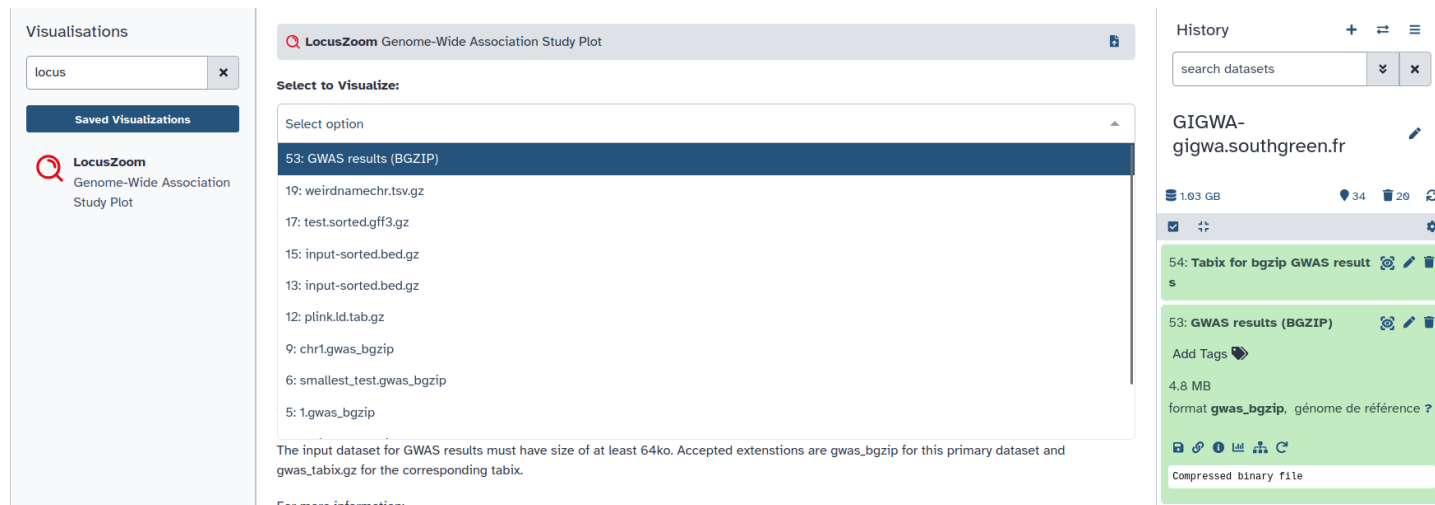


FIGURE 9 – Launching LocusZoom

In the left navigation bar, click on *Visualizations* and search for LocusZoom, then choose the **BGZIP** file (second-to-last generated file).



FIGURE 10 – Launching LocusZoom

Select the **Tabix** file (associated with the **BGZIP** file) and use the following search parameters : chromosome, start, and end. The other parameters are pre-configured to work with the pipeline output files, but they can be changed if the files were generated using a different procedure.